

Counterparty Credit Risk Engine

Final Presentation: Methods, Trade-offs, and Results

Berkeley MFE Industry Project – Capitolis

September 30, 2026

Agenda

Project at a Glance

Engine Architecture

Stochastic Model Choices

Simulation Methodology

Headline Results

Sensitivities (Greeks)

Stress Testing

Model Validation and Model Risk

Credit and Capital: xVA

Conclusions

16 trades, 3 counterparties

- 8 equity total-return swaps
- 4 bond forwards, 4 bond total-return swaps
- 37 equities (US + Tokyo) + USDJPY

A full Monte Carlo engine, built from scratch

- Joint stochastic simulation of every risk factor
- Exposure, Greeks, stress, CVA/DVA/FVA/KVA, SA-CVA, SA-CCR
- 168 automated regression tests

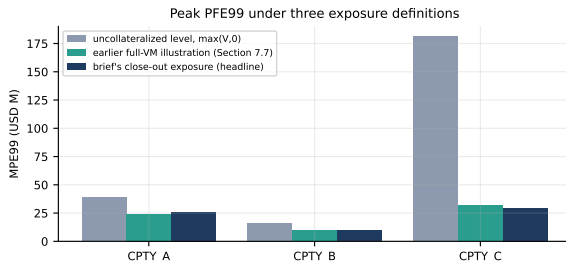
Every input is real, sourced data

- Equities/dividends: Yahoo Finance
- USD curve: real Databento SOFR futures
- Treasury yields: FRED
- Long end + vol cube: licensed Bloomberg data
- JPY curve: real JPY OIS history

Exposure definition, and why two numbers

Close-out: $\max(V(t+10bd) - V(t-1bd), 0)$ Level: $\max(V(t), 0)$

- **Close-out** (the brief's definition, primary): the 10-day move from an already-margined start
- **Level** (secondary): whole mark-to-market, no margin assumed – a no-margin upper bound
- Real answer sits between the two; the actual CSA's threshold/MTA/IM were not supplied



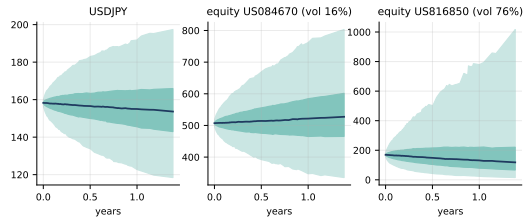
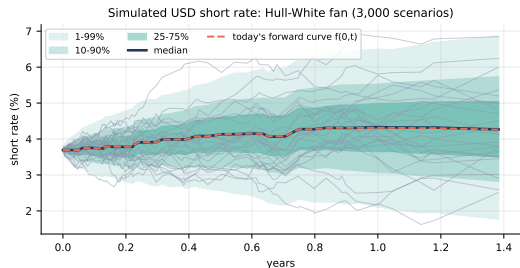
The simulation pipeline

1. **Draw:** Latin Hypercube random numbers $\rightarrow$ correlated shocks (Cholesky)
2. **Simulate:** step every risk factor jointly across the full time grid
3. **Reprice:** feed simulated market states into the provided pricers, every trade, every date, every scenario
4. **Aggregate:** net by counterparty $\rightarrow$ EE / median PFE / PFE99 / MPE
5. **Downstream:** Greeks, stress, CVA/DVA/FVA/KVA, SA-CVA, SA-CCR – all on the same simulated paths

Risk factors simulated jointly, every scenario:

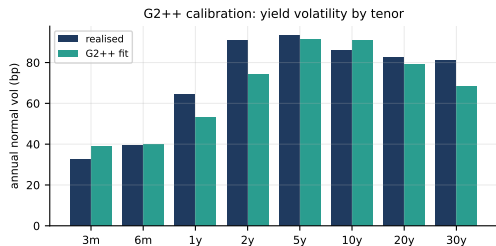
- 37 equities + USDJPY: correlated geometric Brownian motion
- USD short rate: one-factor Hull-White
- JPY short rate: a second, independent Hull-White factor

What the simulated paths look like



Left: simulated USD short rate (median tracks today's forward curve, since the model is fitted to it). Right: simulated USDJPY and two equities with percentile bands.

USD rates: one-factor vs. two-factor Hull-White

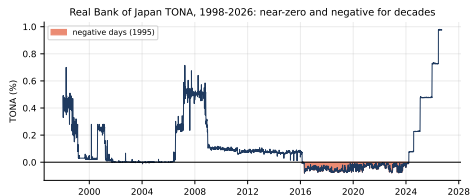


	1-factor	2-factor (G2++)
Fit to realised yield covariance	–	16.1% error
Parameter at a bound	no	$b = 1.5$, $\rho = -0.95$
Effect on portfolio baseline MPE99		–3%

Selected: the two-factor fit strains to match a hump it

can't represent, and moves the headline number by only 3%. The book's rate risk is one big level bet (a single long bond), not a curve-shape bet.

JPY rates: a dedicated factor vs. a constant differential

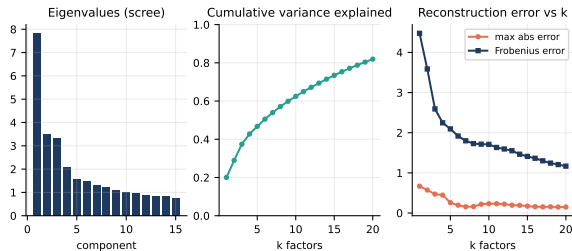


Alternative rejected: JPY names/USDJPY drift off one constant USD-JPY rate differential – cheap, but cannot represent JPY rates moving independently, or negative.

Selected: a second, real, simulated Hull-White factor from the JPY OIS curve.

- TONA has been negative or near-zero for most of 25 years – a floored model can't represent this
- Day-to-day USD/JPY rate changes are uncorrelated (-0.04) despite yield *levels* moving together (~ 0.38)
- Moved portfolio MPE99 attribution from \$52.1M to \$51.5M – real, disclosed, not noise

Equity/FX correlation: full matrix vs. PCA factor model



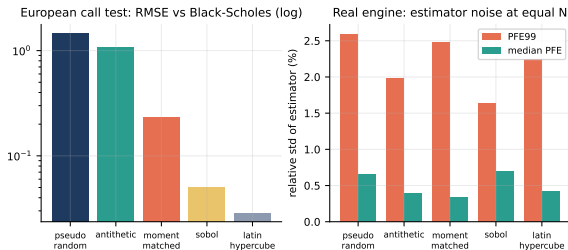
	Full	PCA k=5
Step time	1.91s	2.08s
Variance explained	100%	47%

Selected full matrix: PCA is *slower*, not faster, at $n=39$ (draws k systematic + n idiosyncratic, no dimensionality win). PCA kept as an explainability tool only – factor 1 is a broad market factor, factor 2 is Japan vs. US.

Calibration: mean reversion, and implied vs. realised vol

Parameter	Source used	Why
Rate mean reversion (a)	Real USD swaption cube (implied), $a=0.0167$	Standard route; decay shape is what implied vol reveals. Futures-decay cross-check gives 0.0458 (different instrument/tenor range)
Rate volatility (σ)	Realised long-end Treasury vol, 0.96%	Overnight-SOFR vol (0.63%) understates long-end moves this book depends on; confirmed by Kupiec backtest
Equity/FX volatility	Realised, 3y history	No implied-vol source available; disclosed proxy
JPY mean reversion	Lowest admissible ($a=0.001$)	All 3 real routes (swaption, JGB, OIS) give a <i>negative</i> a – JPY vol rises with tenor

Sampling scheme: five methods compared



Tested: pseudo-random, antithetic, moment-matching, Sobol, Latin Hypercube – on a controlled Black-Scholes benchmark and the real 663-dimensional engine. **Selected** Latin

Hypercube: ~51x lower error on the controlled test, and the only method better than pseudo-random on *both* real-engine statistics (PFE99 and median PFE), at negligible extra cost.

Scenario count vs. precision

N	Rel. SE, PFE99	Time (8 cores)	Use
1,000	higher	~2 min	Iteration / what-if
5,000	moderate	~8 min	Selected Standard reporting
10,000	lower	~15 min	Limit sign-off
15,000+	diminishing returns	–	Not recommended

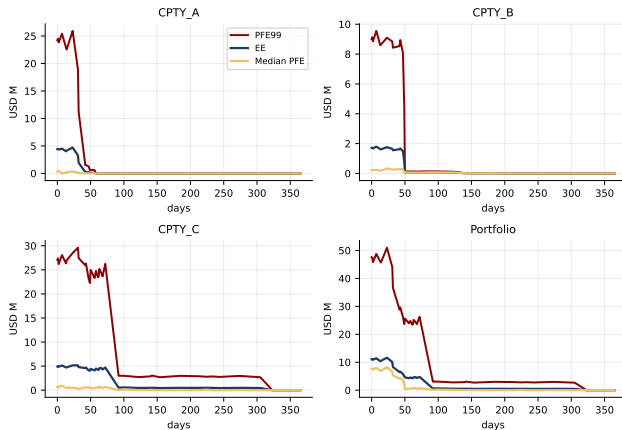
Beyond ~15,000 scenarios, remaining sampling error is an order of magnitude smaller than model-risk

sensitivities (Section 7) – more scenarios stop being the binding constraint on precision well before that.

Headline results (5,000 scenarios, close-out exposure)

	MPE99	Peak EE
CPTY_A	\$25.9M	\$4.7M
CPTY_B	\$9.6M	\$1.8M
CPTY_C	\$29.6M	\$5.2M
Portfolio	\$51.0M	\$11.6M

- CVA \$11.7k close-out / \$247k level
- SA-CVA capital \$0.10M / \$2.23M
- SA-CCR EAD \$260M
- KVA (10% CoC) \$22.4k / \$470k



What drives each counterparty, and why the portfolio is close to additive

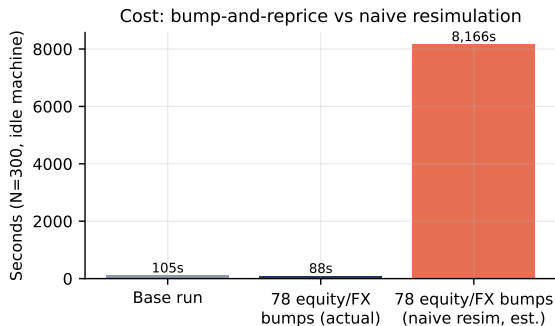
	Equity $\Delta/+1\%$	DV01/+1bp	Rank
CPTY_A	-\$2.25M	\$31k	
CPTY_B	-\$0.96M	-\$4k	
CPTY_C	-\$1.24M	\$594k	

correlation of exposures: A-B 0.34, A-C 0.33, B-C 0.12

- A is most equity-driven; C is the rate netting set (one \$500M short forward on a 2049 Treasury)
- Yield-equity correlation ≈ 0.00 in our data – no reliable offset
- Nothing nets *across* counterparties by construction, and all three peak in the same weeks
- Portfolio MPE99 (\$51.0M) $\approx$ A+C (\$54.4M)

Method, and the cost of a naive approach

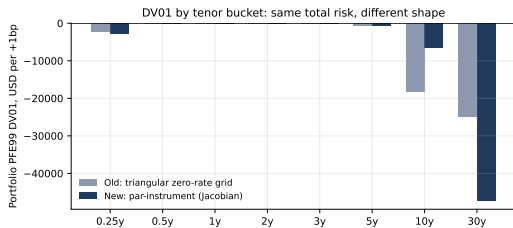
- Bump-and-reprice with **common random numbers**: same draws base/bumped $\rightarrow$ noise \$245k $\rightarrow$ \$1k (**282x lower**)
- Equity/FX bumps rescale simulated GBM paths exactly – no resimulation
- Rate/vol bumps need a genuine resimulation



Bump-and-reprice vs. pathwise vs. adjoint

Method	Cost	Verdict
Bump, independent draws	baseline, very noisy	rejected: noise swamps the signal
Bump, common random numbers	~260s (equity EE)	Selected: works on black-box pricers, works for quantiles
Pathwise (closed form)	0.2–0.4s	Selected for equity/FX only: ~1000x faster, validated to median 0.02%/worst 0.4% vs. bump-and-reprice
Adjoint differentiation	would price all Greeks for ~3-5 valuations	rejected: needs pricers rewritten in differentiable form – out of scope

Rate DV01: zero-grid bump vs. par-instrument Jacobian



Old: bump the *fitted* zero curve at 8 hand-picked tenors, triangular interpolation.

Selected New: bump the curve's own ~45 native construction pillars (real SOFR futures + Bloomberg points), grouped into the same 8 buckets – the genuine par-instrument (Jacobian) sensitivity.

- 30y bucket: **82%** of DV01 (new) vs. 54% (old)
- Old grid smeared the 20–50y risk, where the 2049 bond forward lives, into the 10y bucket
- Both sum to the independent parallel bump within 0.1%

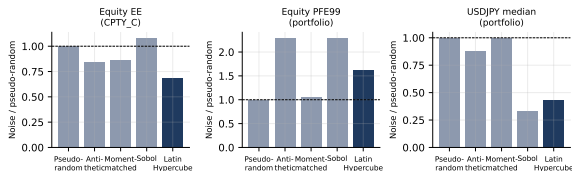
An exact rate Greek, reusing the same simulated draws

In one-factor Hull-White, $r(t) = x(t) + \alpha(t)$: $x(t)$ is the simulated random factor; $\alpha(t)$ is deterministic, depends only on the curve. A curve bump (fixed a, σ) leaves $x(t)$ **identical** to the base run.

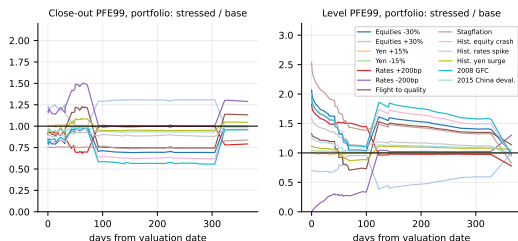
- The whole bump effect is one closed-form number per date, added to the base run's own paths
- **No resimulation. No new random draws.**
- Validated against a true independent resimulation: agreement to 10^{-15} relative precision

Separately: does Latin Hypercube itself reduce Greeks noise? Tested directly – mixed result:

Sampling scheme vs Greeks noise: no consistent winner



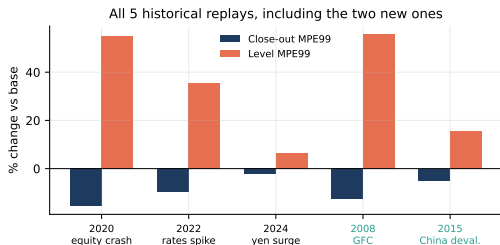
Design: 13 fixed scenarios, same random draws as base



- 8 hypothetical: equities $\pm 30\%$, yen $\pm 15\%$, rates $\pm 200\text{bp}$, flight-to-quality, stagflation
- 5 historical replays, chosen by an objective worst-window rule, not by hand
- Read at every date, every trade, every counterparty

Close-out MPE99 moves at most -24% to $+23\%$; level moves up to $+89\%$ – margin is why. Equity swaps react only to equity/FX; bond trades react only to rates.

Extending the history: 2008 and 2015



Extended public price history back to 2007 (28 of 37 names covered), found the worst 10-day window inside two named crises:

- **2008 GFC:** Sep 29 – Oct 10, 2008; –23% median / –44% worst single-name
- **2015 China deval.:** Aug 2015, –6.8% median

Result: both land *inside* the range the existing scenarios already covered – a validation finding, not just more coverage.

Four independent validation checks

Check		Method	Result
Martingale test		Discounted paths average to today's price	Pass
Parametric benchmark	VaR	Delta-normal VaR vs. full MC P&L	\$11.4M vs. \$11.7M (ratio 0.98)
Kupiec (99%)	backtest	Historical exceedances vs. prediction	4/6/2 exceptions vs. 2.4 expected – all pass
Excel checks	parametric	Independent spreadsheet formulas, public data	0.1–7% agreement across 4 checks

The Excel workbook recomputes calibrated vols live ($=STDEV(LN(P_t/P_{t-1}))*SQRT(252)$), the simulated vol

against the closed-form $\text{vol} \times \sqrt{T}$, and a Monte Carlo vega against the closed-form Black vega under a $\pm 1\%$ bump – every number a live formula on public data, no code trust required.

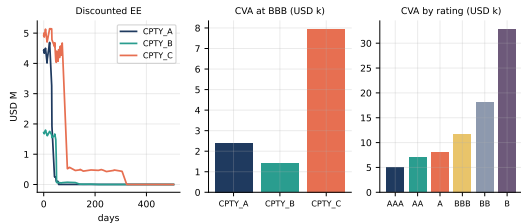
Model risk: what actually moves the headline number

Perturbation (base portfolio MPE99 = \$51.5M)	Change
Equity/FX vol $\times 2$	+71%
Equity correlations, 30% toward 1	+22%
Equity/FX vol $\times 1.25$	+19%
Rate–equity dependence, both fall together	+18%
Rate volatility $\times 1.25$	+9%
Rate–equity dependence, offsetting	–15%
Mean reversion $a \times 3$	–5%
Two-factor rate model (G2++) instead of one-factor	–3%

Conclusion: equity volatility, correlation,

and the data-unresolvable rate-equity dependence dominate. The one- vs. two-factor rate-model choice explored in depth is, by comparison, one of the *smallest* levers – the strongest argument for the simpler model.

CVA: methodology and results

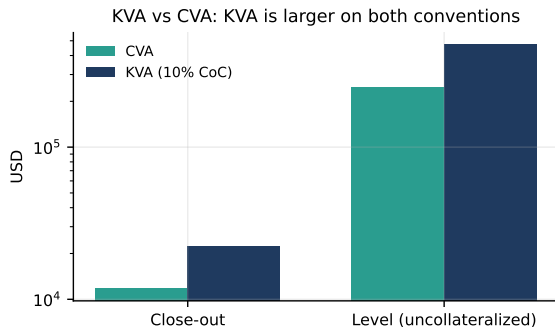


- No CDS available → ICE BofA bond-index OAS proxy by rating (Basel MAR50.32(3)-compliant)
- Credit-triangle hazard-rate survival model:
 $\text{hazard} \approx \text{spread} / (1 - \text{recovery})$
- LGD 60%; all three counterparties proxied at BBB

Value delivered: the sensitivity chain – CVA01, the rating table (AAA \$5.0k → B \$32.8k), SA-CVA weighted sensitivities by risk class – matters more than the point-in-time number itself.

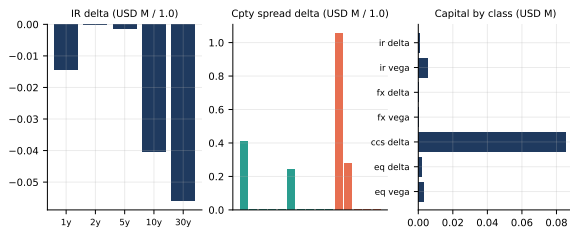
DVA, FVA, and KVA (extra credit)

	Close-out	Level
CVA	\$11.7k	\$247k
DVA	\$11.2k	\$7.5k
FVA	\$0.5k	\$240.0k
KVA (10% CoC)	\$22.4k	\$470k



KVA: $K(t) = 8\% \times RW \times \alpha \times EPE(t)$, cost-of-capital swept 8/10/12% (none given). Exceeds CVA because it scales the whole EPE profile by a fixed ratio, not a small IG default probability – explicitly illustrative.

Regulatory capital: SA-CVA and SA-CCR



- SA-CVA capital: \$0.10M close-out / \$2.23M level (RWA \$1.2M / \$27.8M)
- Dominated by counterparty credit-spread risk in both conventions
- SA-CCR exposure at default: \$260M, dominated by the replacement cost of BF_0003

The kickoff brief flagged “*Risk bonds + CDS data for new sample trades*”. Delivered:

- A sample risky-bond forward (25M notional, 5% BBB corporate; not part of the ESF book)
- Priced and simulated with a real issuer credit curve, re-anchored at every simulated node
- Same hazard-rate survival model and bond-index proxy construction as the counterparty CVA

Limitation, disclosed: no real CDS quotes were obtainable for a hypothetical issuer, so the issuer curve is a bond-index proxy, not actual CDS – delivered in spirit, not the full ask.

Recap: every trade-off, one line each

Decision	Why
One-factor Hull-White, not two-factor	Two-factor moves MPE99 by only -3% ; 16% fit error; book is a level bet
Dedicated JPY Hull-White factor	Real, independent JPY rate risk incl. negative rates; moved MPE99 \$52.1M $\rightarrow$ \$51.5M
Full correlation matrix, not PCA	PCA is slower at $n=39$, no accuracy gain; kept as explainability tool only
Latin Hypercube sampling	Best of 5 schemes on controlled & real-engine tests; no consistent Greeks-noise edge
Close-out exposure primary, level secondary	The brief's definition; level is an honest no-margin upper bound
Bump-and-reprice with CRN, pathwise cross-check	282x noise reduction; pathwise 1000x faster for linear risk
Par-instrument DV01 buckets	Corrected real mis-attribution: 82% vs. 54% of risk in the 30y bucket
Exact LHS-reuse rate Greek	Eliminates resimulation cost for curve-only bumps, 10^{-15} validated
Rating-proxy CVA/DVA/FVA/KVA	No CDS/real ratings available; Basel-compliant proxy, fully disclosed

- Keep the one-factor Hull-White rate model; revisit only if curve-shape risk grows materially
- Keep the full correlation matrix; retain PCA as an explainability exhibit
- Keep Latin Hypercube as the default sampling scheme
- Adopt the par-instrument DV01 buckets as the primary rate-risk breakdown
- Treat CVA/DVA/FVA/KVA as sensitivity-generating tools first, point estimates second

Open items for the next phase: real counterparty ratings or CDS, Capitolis' own credit/funding spread, a real capital methodology and cost of capital for KVA, a decision on wrong-way risk, and the true CSA terms (threshold, MTA, IM).

Thank you

Questions and discussion

Appendix

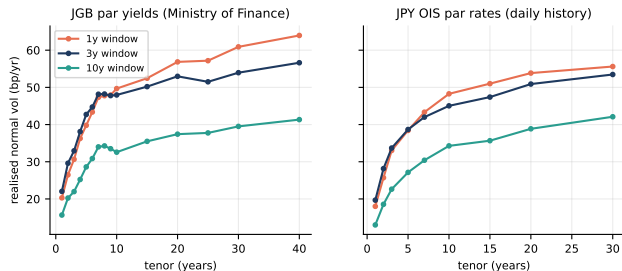
Backup detail

Appendix: correction attribution, one at a time

Step	Portfolio MPE99
Earlier version (flat curve beyond 6.5y, SOFR-overnight σ , no JPY factor, USDJPY drift sign error)	\$45.7M
+ Corrected USDJPY drift sign	\$47.8M
+ Long-end volatility fit and 10y-yield correlations	\$52.7M
+ Bloomberg long end spliced onto the USD curve	\$52.1M
+ JPY Hull-White factor (final model)	\$51.5M

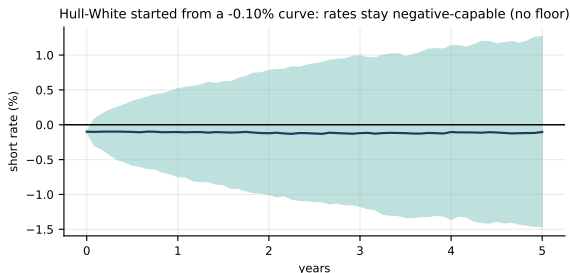
Appendix: JPY mean-reversion calibration routes

All three real-data routes for JPY give a negative a (vol rises with tenor, the reverse of what Hull-White assumes), so the Ho-Lee lower bound ($a=0.001$) is used instead of an invalid fit.

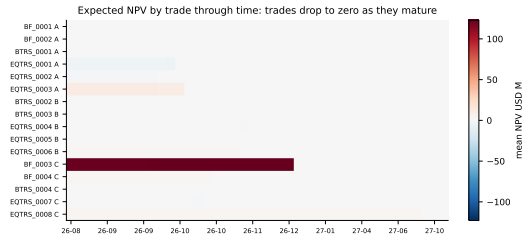


Appendix: negative-rate capability, synthetic demonstration

Our only real JPY curve snapshot (Aug 2026) is positive after BOJ hikes, so the negative case is demonstrated synthetically: Hull-White started from a synthetic -0.10% flat curve.

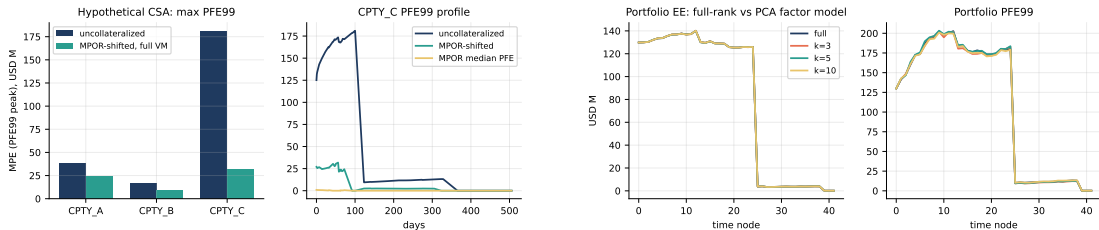


Appendix: exposure diagnostics



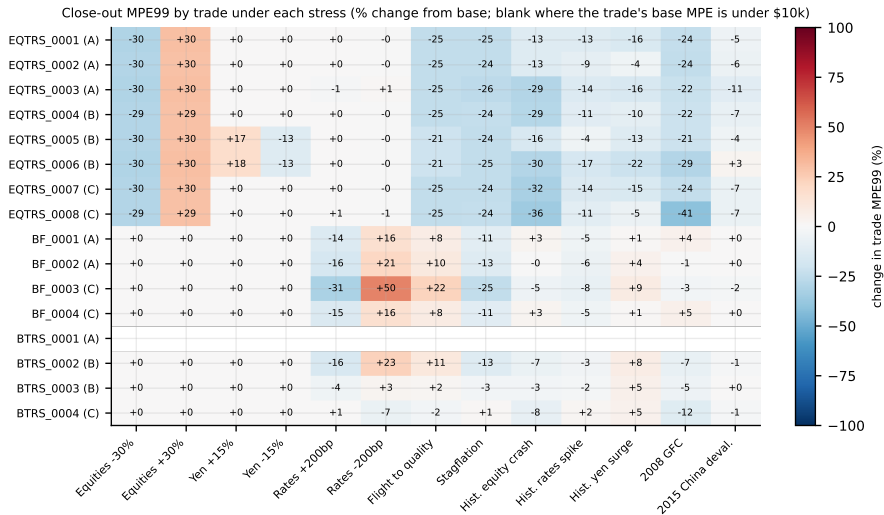
Left: probability of positive exposure by counterparty. Right: expected NPV by trade – BF_0003 dominates and disappears at its Dec 2026 settlement.

Appendix: full-VM comparison and PCA exposure cross-check

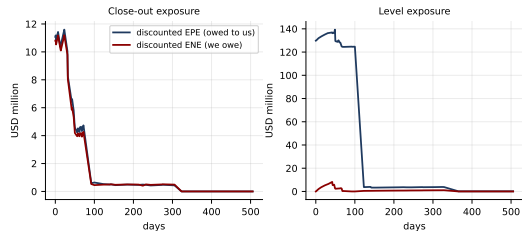
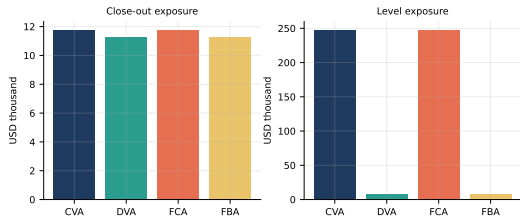


Left: peak PFE99 under a hypothetical full-VM CSA. Right: level exposure under the full correlation matrix vs. a k -factor PCA model – curves coincide closely from $k=5$.

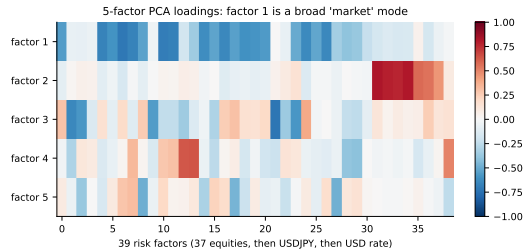
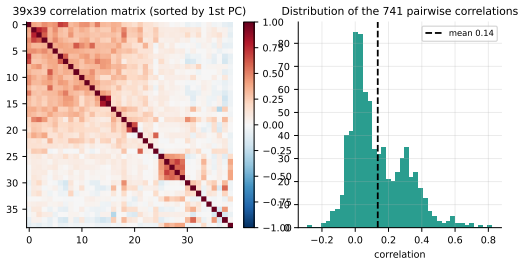
Appendix: per-trade stress sensitivity heat map



Appendix: xVA components and discounted exposure



Appendix: correlation matrix and factor loadings



Appendix: feedback tracker (both rounds)

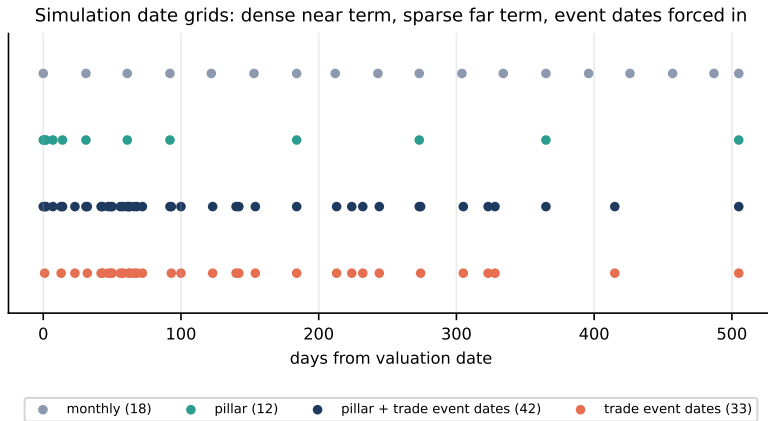
Earlier meeting

1. Robust stress testing **Selected**
2. Sensitivities: method, cost, faster alt. **Selected**
3. PCA explainability and speed **Selected**
4. Final results explanation **Selected**
5. CVA walkthrough and set-up **Selected**

This meeting

6. 2008 and more testing periods **Selected**
7. LHS to calculate sensitivities **Selected**
8. KVA addition **Selected**
9. CVA is for sensitivities **Selected**
10. DV01 via a Jacobian **Selected**
11. Excel parametric vol-bump test **Selected**

Appendix: simulation grid design



The pillar grid (12 dates) plus forced trade-event dates (42 total) is more compact than a plain monthly grid (18) and hits every real cash-flow date exactly.